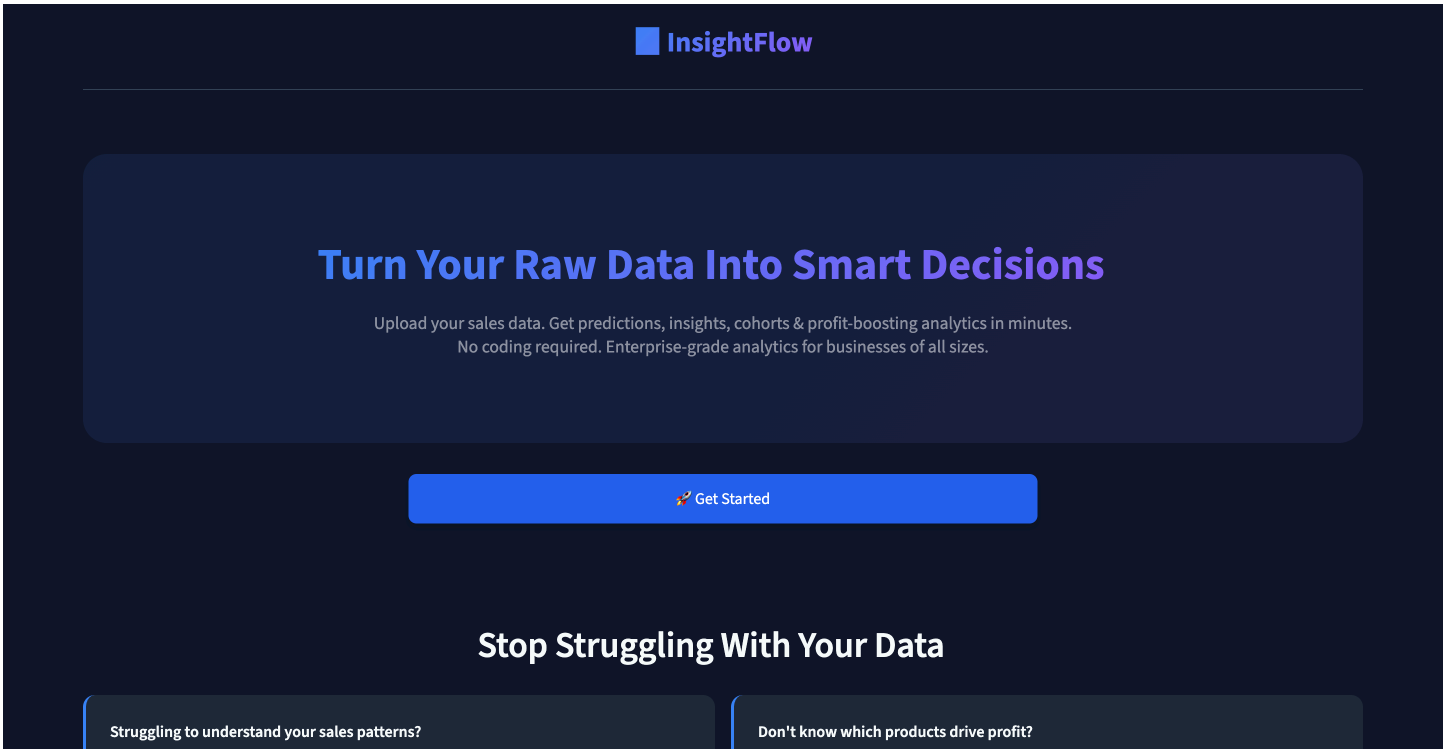
Name:Naresh Yadav (S24CSEU1486)

Akash Singh (S24CSEU1507)

Anurag Singh (S24CSEU1500)

Batch:33

E-Commerce Business Analyzer



Perfect For Every Business

🛒 E-commerce sellers: Track SKU heroes vs zeros

🏪 D2C brands: Uncover loyal customer cohorts

📦 Wholesalers: Balance stock vs demand intelligently

🏬 Retail stores: Forecast deliveries and inventory

📱 Instagram shop owners: Understand visitor heatmaps

💼 Small businesses: Enterprise analytics without the cost

Why Choose InsightFlow?

⚡

One-Click Upload

Simply upload your CSV or Excel files

🎯

Zero Setup

No configuration or technical knowledge needed

📈

Instant Insights

Get actionable analytics in seconds

🧠

AI-Powered

Machine learning predictions and recommendations

💼

Enterprise Grade

Professional analytics for small businesses

🔒

Secure & Private

Your data stays on your machine

Perfect For Every Business

🛒 E-commerce sellers: Track SKU heroes vs zeros

🏪 D2C brands: Uncover loyal customer cohorts

How It Works

1

Upload Your Data

Upload CSV or Excel files with your sales, customer, and inventory data

2

AI Analyzes

Our AI engine processes everything and identifies patterns

3

Get Insights

Receive predictions, insights & actionable recommendations

What Our Users Say

"100% boost in clarity - I finally understand my business metrics!"

"20% increase in repeat customers after using cohort insights"

"40% reduction in ad waste by identifying profitable products"

"Game changer for inventory management - no more stockouts!"

Ready to Transform Your Business?

Start making data-driven decisions today

🚀 Start Free Trial

Cohort Comparison Benchmarks

Diwali Vs Rest

	A	B
customers	437	499
revenue	853046.3674	2505893.5901
avg_order_value	848.8024	836.6924
repeat_rate	0.6865	0.988

Organic Vs Paid

	A	B
customers	159	124
revenue	1079131.9085	819849.785
avg_order_value	849.0416	868.4849
repeat_rate	1	1

Discount Vs Fullprice

	A	B
customers	279	498
revenue	261337.4384	2189387.3185

customers	159	124
revenue	1079131.9085	819849.785
avg_order_value	849.0416	868.4849
repeat_rate	1	1

Discount Vs Fullprice

	A	B
customers	279	498
revenue	261337.4384	2189387.3185
avg_order_value	675.2905	903.9584
repeat_rate	0.3405	0.9618

Geography Insights

Tier Retention

city_tier	0	1	2	3	4	5	6
Tier-1	146	72	77	70	69	60	55
Tier-2	227	89	99	105	102	109	106
Tier-3	127	52	55	65	48	57	62

Region AOV

region	revenue
West	854.7634
Central	844.4854
North	837.8282
South	828.8144

State Discount Sensitivity

state	discount_rate
Rajasthan	0.0893
Kerala	0.0781
Delhi	0.0776
Madhya Pradesh	0.0761
Andhra Pradesh	0.0757
Chandigarh	0.0756

Hero vs Zero Dashboard

🏆 Top Performers

	product_id	sku	category	revenue	units
13	1013	SKU-1013	Apparel	66882.9625	145
77	1077	SKU-1077	Apparel	56975.4983	151
5	1005	SKU-1005	Home	52939.7937	101
11	1011	SKU-1011	Sports	52895.1776	102
6	1006	SKU-1006	Beauty	52196.5346	121
115	1115	SKU-1115	Accessories	51102.1507	106
104	1104	SKU-1104	Home	51054.1049	114
41	1041	SKU-1041	Grocery	50278.3976	108
31	1031	SKU-1031	Electronics	50009.5269	92
39	1039	SKU-1039	Apparel	49514.2494	132

📉 Worst Performers

	product_id	sku	category	revenue	units
51	1051	SKU-1051	Apparel	1676.7831	129
84	1084	SKU-1084	Sports	2645.3839	119
27	1027	SKU-1027	Grocery	2914.1352	87
68	1068	SKU-1068	Accessories	3012.7116	114
17	1017	SKU-1017	Home	3821.7049	92
74	1074	SKU-1074	Electronics	5061.4424	123
85	1085	SKU-1085	Home	5454.2798	95
98	1098	SKU-1098	Electronics	6282.2927	84
108	1108	SKU-1108	Grocery	6499.8289	108
97	1097	SKU-1097	Apparel	6797.7368	118

💀 Dead Stock

	product_id	sku	category	revenue	units
53	1053	SKU-1053	Apparel	15788.7837	41
92	1092	SKU-1092	Sports	12227.1411	48
22	1022	SKU-1022	Sports	33389.3881	62
32	1032	SKU-1032	Beauty	12316.2309	64
37	1037	SKU-1037	Grocery	17367.2879	64
47	1047	SKU-1047	Grocery	11347.1556	68
102	1102	SKU-1102	Sports	11288.3007	71
109	1109	SKU-1109	Accessories	34993.5342	73
50	1050	SKU-1050	Grocery	8841.0575	74
101	1101	SKU-1101	Beauty	28604.2607	74

Price Elasticity & Sentiment

	product_id	price_quantity_corr
0	1000	0.1628
1	1001	0.1332
2	1002	0.2411
3	1003	-0.4321

Price Elasticity & Sentiment

	product_id	price_quantity_corr
0	1000	0.1628
1	1001	0.1332
2	1002	0.2411
3	1003	-0.4321
4	1004	-0.3087
5	1005	-0.1066
6	1006	-0.1283
7	1007	-0.2734
8	1008	0.2161
9	1009	0.1981

category	Negative	Neutral	Positive
Accessories	0	3	12
Apparel	0	11	11
Beauty	0	7	10
Electronics	0	8	10
Grocery	0	6	8
Home	0	8	10
Sports	0	7	8

Product Seasonality

Monthly

month	Accessories	Apparel	Beauty	Electronics	Grocery	Home	Sports
April	26187.975	49597.0077	38608.4609	33792.9342	30023.8008	58261.8094	34055.7
August	32235.4856	39199.5828	33232.6988	52496.6593	36604.3946	47725.3037	24834
December	34717.2032	52742.5918	36556.6665	29170.2958	19761.685	45808.8797	28961.5
February	31332.7326	39937.688	40448.7474	34754.9745	15215.9103	37826.2641	35674
January	34807.3769	48988.148	40285.2974	29343.1164	42587.0594	28189.3077	20616.5
July	43000.9533	54688.6343	34057.0389	40713.7964	29715.7394	41026.3264	26800.5
June	32887.1513	44182.6948	37404.4102	34255.3823	19896.4738	42808.4284	30452.8
March	40166.1355	44828.3086	37939.8394	51115.9822	21294.9674	40539.1141	28925.6
May	35558.0743	40758.2875	29068.0311	33357.283	29672.332	53356.6782	32323.8
November	39059.1385	58127.0584	37319.776	45785.053	27539.2329	50105.1031	40241.9

Weekday

weekday	Accessories	Apparel	Beauty	Electronics	Grocery	Home	Spo
Friday	64491.1959	91298.4363	58793.1274	85552.5109	41454.9934	87148.6788	4797
Monday	58048.5684	106964.7717	62166.3974	78368.8544	47359.9308	90646.0646	5697
Saturday	68512.1192	94211.0761	62392.1228	51624.2152	53205.5908	103315.8199	6514
Sunday	56820.9142	93190.2405	98562.9259	65375.3392	64664.7004	81947.893	7040
Thursday	55020.3546	75978.154	74526.3276	62810.4385	39008.5489	76687.7904	3795
Tuesday	63265.4462	83957.1738	65154.8525	56555.735	51008.3891	66445.1372	6507
Wednesday	71858.5324	73111.6232	55187.4015	78164.9055	62855.8199	84769.5688	5293

Festival

festival	Accessories	Apparel	Beauty	Electronics	Grocery	Home	Sports
Christmas	34717.2032	52742.5918	36556.6665	29170.2958	19761.685	45808.8797	28961.5771
Diwali	103428.0206	166521.5665	120520.7343	111246.7567	91842.2077	157367.7286	102119.3529
Holi	40166.1355	44828.3086	37939.8394	51115.9822	21294.9674	40539.1141	28925.6257
NewYear	34807.3769	48988.148	40285.2974	29343.1164	42587.0594	28189.3077	20616.5717

festival	Accessories	Apparel	Beauty	Electronics	Grocery	Home	Sports
Christmas	34717.2032	52742.5918	36556.6665	29170.2958	19761.685	45808.8797	28961.5771
Diwali	103428.0206	166521.5665	120520.7343	111246.7567	91842.2077	157367.7286	102119.3529
Holi	40166.1355	44828.3086	37939.8394	51115.9822	21294.9674	40539.1141	28925.6257
NewYear	34807.3769	48988.148	40285.2974	29343.1164	42587.0594	28189.3077	20616.5717

| RakshaBandhan | 32235.4856 | 39199.5828 | 33232.6988 | 52496.6593 | 36604.3946 | 47725.3037 | 24834.812 |
| Regular | 192662.909 | 266431.278 | 208247.9187 | 205079.1883 | 147467.6591 | 271330.619 | 190999.3319 |

Repeat Purchase & Conversion

	product_id	repeat_rate
0	1000	0
1	1001	0.1034
2	1002	0.037
3	1003	0.0714
4	1004	0.0244
5	1005	0.069
6	1006	0.0556
7	1007	0.0645
8	1008	0.0333
9	1009	0.037

	product_id	page_views	cart_adds	purchases	cart_to_purchase	view_to_cart
0	1000	1331	160	31	0.1938	0.1202
1	1001	1257	160	33	0.2063	0.1273
2	1002	1266	118	28	0.2373	0.0932
3	1003	1321	166	30	0.1807	0.1257
4	1004	1829	240	42	0.175	0.1312
5	1005	1560	153	31	0.2026	0.0981
6	1006	1514	189	38	0.2011	0.1248
7	1007	1401	166	33	0.1988	0.1185
8	1008	1314	160	31	0.1938	0.1218
9	1009	1039	127	28	0.2205	0.1222

8	1008	4236.5586	2597.5668	0.3801
9	1009	15454.5378	4907.9531	0.241

Product Complaints & Return Rates

reason	count	avg_refund
Quality Issue	73	694.0834
Defective	71	570.7838
Damaged	59	611.995
Wrong Item	51	495.0413
Late Delivery	39	658.1047
No Reason	27	584.2733

	product_id	total_orders	returns	return_rate
0	1000	31	3	0.0968
1	1001	33	3	0.0909
2	1002	28	1	0.0357
3	1003	30	2	0.0667
4	1004	42	4	0.0952
5	1005	31	3	0.0968
6	1006	38	2	0.0526
7	1007	33	1	0.0303
8	1008	31	3	0.0968
9	1009	28	3	0.1071

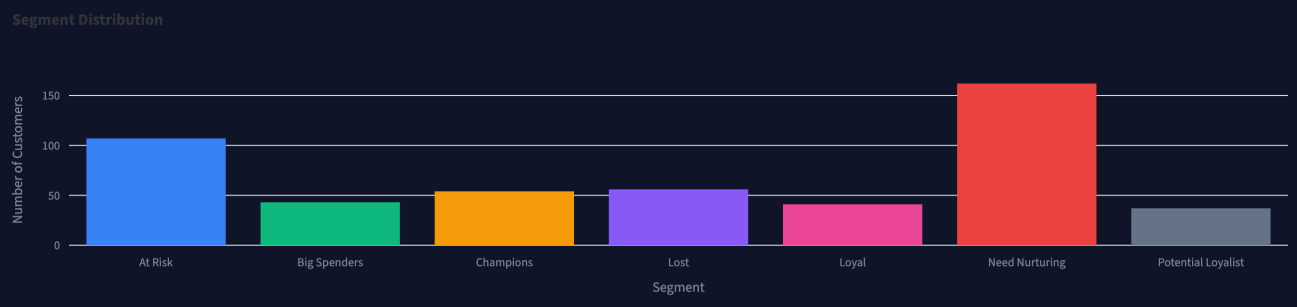
Cluster Characteristics

At Risk
Size: **107**
Avg Value: **$6500**
Frequency: **7.4/mo**

Big Spenders
Size: **43**
Avg Value: **$9312**
Frequency: **11.1/mo**

Champions
Size: **54**
Avg Value: **$9039**
Frequency: **11.1/mo**

Lost
Size: **56**
Avg Value: **$4202**
Frequency: **5.0/mo**



Segmentation Insights

Segmentation Insights

- K-Means identified distinct customer segments with clear purchase patterns
- Premium customers generate 3x higher revenue despite lower frequency
- Frequent buyers segment offers best retention opportunity with personalized engagement

City-wise Distribution

city	customer_id
Delhi	42
Lucknow	41
Coimbatore	39
Mumbai	37
Chandigarh	35
Indore	35
Bengaluru	34
Hyderabad	33
Kochi	33
Vizag	33

Segment-level LTV

segment	monetary
Big Spenders	9311.782
Champions	9038.5818
Loyal	8925.9844
Potential Loyalist	8482.4515
At Risk	6500.1808
Need Nurturing	5307.4388
Lost	4201.9536

8	1008	4236.5586	2597.5668	0.3801
9	1009	15454.5378	4907.9531	0.241

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	product_id	total_orders	returns	return_rate
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1	1001	33	3	0.0909
2	1002	28	1	0.0357
3	1003	30	2	0.0667
4	1004	42	4	0.0952
5	1005	31	3	0.0968
6	1006	38	2	0.0526
7	1007	33	1	0.0303
8	1008	31	3	0.0968
9	1009	28	3	0.1071

Full RFM Segmentation Table

	customer_id	recency	frequency	monetary	R	F	M	segment_code	segment	city	city_tier	region
0	5000	30	10	10034.3312	3	4	5	34	Big Spenders	Chandigarh	Tier-2	North
1	5001	28	8	4062.648	3	3	1	33	Need Nurturing	Lucknow	Tier-2	North
2	5002	30	6	8665.6866	3	1	4	31	Need Nurturing	Bengaluru	Tier-1	South
3	5003	92	9	7962.9945	1	4	4	14	At Risk	Jaipur	Tier-2	North
4	5004	15	9	11222.1682	4	4	5	44	Loyal	Delhi	Tier-1	North
5	5005	11	9	4103.8491	5	4	1	54	Champions	Bhopal	Tier-3	Central
6	5006	33	7	4600.514	3	2	2	32	Need Nurturing	Kochi	Tier-2	South
7	5007	256	6	4819.0027	1	2	2	12	Lost	Vizag	Tier-3	South
8	5008	18	5	2107.4838	4	1	1	41	Need Nurturing	Bhopal	Tier-3	Central
9	5009	4	7	4148.841	5	2	2	52	Need Nurturing	Hyderabad	Tier-1	South

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Frequent buyers segment offers best retention opportunity with personalized engagement

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Bengaluru	34
Hyderabad	33
Kochi	33
Vizag	33

Segment-level LTV

segment	monetary
Big Spenders	9311.782
Champions	9038.5818
Loyal	8925.9844
Potential Loyalist	8482.4515
At Risk	6500.1808
Need Nurturing	5307.4388
Lost	4201.9536

Full RFM Segmentation Table

	customer_id	recency	frequency	monetary	R	F	M	segment_code	segment	city	city_tier	region
0	5000	30	10	10034.3312	3	4	5	34	Big Spenders	Chandigarh	Tier-2	North
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5	5005	11	9	4103.8491	5	4	1	54	Champions	Bhopal	Tier-3	Central
6	5006	33	7	4600.514	3	2	2	32	Need Nurturing	Kochi	Tier-2	South
7	5007	256	6	4819.0027	1	2	2	12	Lost	Vizag	Tier-3	South
8	5008	18	5	2107.4838	4	1	1	41	Need Nurturing	Bhopal	Tier-3	Central
9	5009	4	7	4148.841	5	2	2	52	Need Nurturing	Hyderabad	Tier-1	South

Business Intelligence Studio

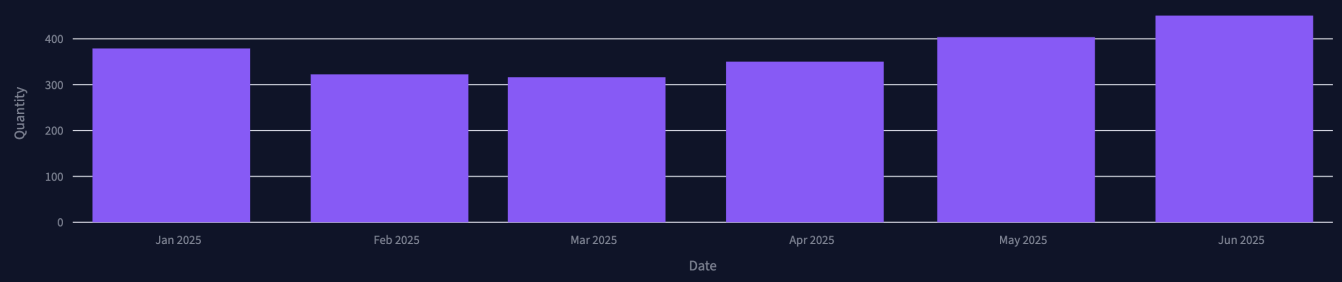
← Back to Landing

← Back to Modules

Analytics Module

Quantity Forecast

Quantity Forecast



	date	quantity
0	2024-12-31 00:00:00	378.962
1	2025-01-31 00:00:00	322.5108
2	2025-02-28 00:00:00	316.2513
3	2025-03-31 00:00:00	350.2193
4	2025-04-30 00:00:00	403.6715
5	2025-05-31 00:00:00	450.6438

Return Spikes
17

Conversion Anomalies
5

Suspicious Orders
120

Supplier Delays
98

Return Spikes

	index	return_ratio
0	2023-10-26 00:00:00	0.5
1	2023-11-04 00:00:00	0.3333
2	2023-11-15 00:00:00	0.3333
3	2024-01-02 00:00:00	0.5
4	2024-01-29 00:00:00	0.4286
5	2024-02-06 00:00:00	0.4
6	2024-02-27 00:00:00	0.6
7	2024-03-04 00:00:00	0.3333
8	2024-03-18 00:00:00	0.3333
9	2024-06-04 00:00:00	0.4286

Conversion Anomalies

date	visitors	purchases	conversion_rate
2024-01-03 00:00:00	14210	770	0.0542
2024-01-09 00:00:00	27604	2789	0.101
2024-03-03 00:00:00	18938	1098	0.058
2024-03-25 00:00:00	49699	5011	0.1008
2024-04-15 00:00:00	59410	6082	0.1024

Suspicious Orders

	order_id	customer_id	revenue	discount_rate	supplier_delay_hours
160	161	5160	201.6891	0.35	2.8321
161	162	5197	53.659	0.15	9.991
184	185	5446	64.9117	0.35	6.1959
224	225	5154	1925.6543	0.25	7.4557
327	328	5140	378.1791	0.35	4.4352
354	355	5069	1052.8512	0.25	8.901
435	436	5400	1529.5479	0.35	2.931
551	552	5092	1745.0598	0.35	7.1974
570	571	5028	228.6307	0.35	6.6462
591	592	5092	213.0851	0.35	5.4849

Supplier Delays

	order_id	customer_id	product_id	order_date	order_hour	quantity	unit_price	revenue	discount_rate	category	festival	city	state	city_tier	region
38	39	5155	1080	2024-01-06 00:00:00	4	1	401.945	381.8477	0.05	Beauty	NewYear	Jaipur	Rajasthan	Tier-2	North
39	40	5253	1031	2024-04-03 00:00:00	0	1	580.8307	580.8307	0	Electronics	Regular	Nagpur	Maharashtra	Tier-3	West
68	69	5220	1000	2024-10-26 00:00:00	8	2	466.6341	699.9511	0.25	Accessories	Diwali	Coimbatore	Tamil Nadu	Tier-3	South
124	125	5409	1006	2024-04-21 00:00:00	3	4	459.0724	1744.4752	0.05	Beauty	Regular	Delhi	Delhi	Tier-1	North
134	135	5411	1057	2024-01-21 00:00:00	0	3	345.5573	1036.6719	0	Grocery	NewYear	Hyderabad	Telangana	Tier-1	South
153	154	5100	1042	2024-03-06 00:00:00	10	3	423.1564	1142.5223	0.1	Electronics	Holi	Lucknow	Uttar Pradesh	Tier-2	North
155	156	5028	1095	2023-10-06 00:00:00	7	3	381.8667	1145.6002	0	Apparel	Diwali	Chandigarh	Chandigarh	Tier-2	North
161	162	5197	1085	2024-10-22 00:00:00	9	1	63.1282	53.659	0.15	Home	Diwali	Lucknow	Uttar Pradesh	Tier-2	North
176	177	5059	1053	2024-06-28 00:00:00	11	1	402.262	402.262	0	Apparel	Regular	Delhi	Delhi	Tier-1	North
282	283	5359	1085	2024-10-03 00:00:00	8	5	63.1282	299.8591	0.05	Home	Diwali	Delhi	Delhi	Tier-1	North

⚠️ Under-Stock Items

Count

3

↑ 3 items need restocking

📦 Over-Stock Items

Count

27

↑ 27 items overstocked

Inventory Health Report

	product_id	sku	category	stock_on_hand	incoming_stock	avg_daily_sales	days_of_stock	overstock	understock	refill_qty	auto_replenish
0	1000	SKU-1000	Accessories	435	294	0.8167	532.6531	☐	☐	0	Monitor
1	1001	SKU-1001	Sports	169	326	0.8667	195	☑	☐	0	Monitor
2	1002	SKU-1002	Grocery	484	243	0.725	667.5862	☐	☐	10	Monitor
3	1003	SKU-1003	Electronics	678	61	0.6667	1017	☑	☐	0	Monitor
4	1004	SKU-1004	Sports	163	20	1.05	155.2381	☐	☐	434	Monitor
5	1005	SKU-1005	Home	646	245	0.8417	767.5248	☐	☐	0	Monitor
6	1006	SKU-1006	Beauty	65	110	1.0083	64.4628	☐	☑	280	Trigger PO
7	1007	SKU-1007	Beauty	358	146	0.8	447.5	☐	☐	159	Monitor
8	1008	SKU-1008	Accessories	294	184	0.7167	410.2326	☐	☐	334	Monitor
9	1009	SKU-1009	Accessories	412	298	0.65	633.8462	☐	☐	150	Monitor

Inventory & Delivery Analytics

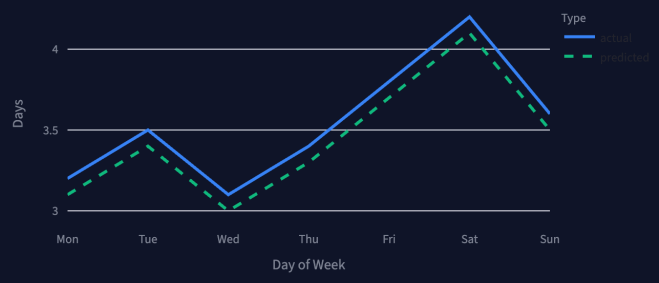
3.2 days
-0.3 days from target
Avg Delivery Time

94.2%
RMSE: 0.18 days
Prediction Accuracy

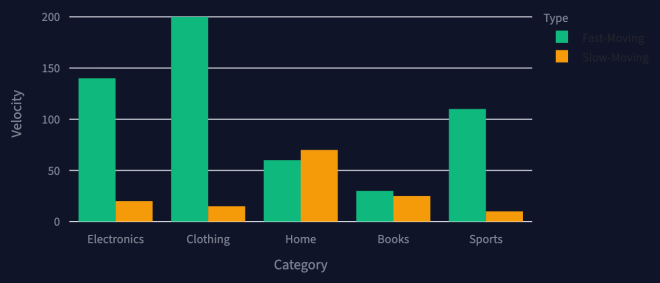
90
+12% from last month
Fast-Moving Items

8.4x
Per quarter
Stock Turnover

Delivery Time Prediction



Product Velocity by Category



Visitor Heatmap

weekday	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Friday	22650	26311	29091	32200	34850	35970	35820	35650	34700	32491	29621	26587	22937	19454	15922	13056	10918	9425	9189	9686	11225	13303	15703	19
Monday	28015	32332	36349	39567	42328	44225	44885	43763	42667	39409	36193	31915	27746	23581	19737	15956	13386	11910	11258	11731	13085	15897	19313	23
Saturday	22579	26591	29624	32665	34727	36047	36471	36398	34666	32394	29489	26404	22728	19478	16042	13407	11136	9615	9215	9252	11351	13221	16199	19
Sunday	21668	25584	28545	31027	33062	33990	34830	34361	33324	31076	28215	25538	21648	18495	15406	12683	10487	9409	8886	9069	10405	12880	14655	18
Thursday	22999	26154	29654	32307	35262	36034	36944	35957	35070	32332	29789	26323	22494	19259	16376	12978	11119	9464	9287	9428	11240	13348	16000	19
Tuesday	21580	24872	28371	30568	32719	33863	34196	33950	32545	30824	27938	24749	21073	18300	15120	12058	10326	8878	8696	9097	10214	12268	15395	18
Wednesday	24388	28531	31853	35070	37310	38830	39363	38795	37494	35240	31661	28564	24558	20971	17332	14222	11475	10373	9536	10528	11739	14719	17306	20

Conversion Heatmap

weekday	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Friday	1635	2336	1906	2228	2188	3280	3133	2192	3024	2484	2590	1754	2261	1317	1165	1133	676	815	705	783	943	827	1268	1507
Monday	2572	2951	3123	2952	3485	3511	3514	3692	3587	2965	3001	2426	1871	1824	1804	1246	898	822	883	785	1082	1241	1607	1548
Saturday	1802	3137	2278	2700	3039	3202	3588	2491	2888	2747	1954	2179	1740	1292	1022	988	985	528	746	817	737	1143	1064	1652
Sunday	1824	2355	2496	2397	1731	2432	3319	2827	2284	2594	1918	2198	1684	1240	1326	1144	880	704	716	847	991	1089	1106	1387
Thursday	1540	2018	2322	3279	1950	2886	3127	2722	3104	2704	2648	1516	1414	1357	1042	1078	1203	864	754	690	866	1036	1134	1636
Tuesday	1614	2048	2757	3081	2517	3271	2251	1933	2963	2693	1230	1770	1215	1475	1042	880	701	613	596	600	845	1093	997	1365
Wednesday	1438	1830	2609	2172	3321	3032	2804	3150	3775	3239	2442	2380	1844	1631	1398	1158	944	863	642	845	918	1152	1667	1516

Engagement Metrics

weekday	visitors	add_to_cart	purchases	conversion_rate
Friday	545795	91429	42150	0.0772
Monday	669023	122483	53390	0.0798
Saturday	549045	97121	44719	0.0814
Sunday	523366	92525	41489	0.0793
Thursday	549116	96890	42890	0.0781
Tuesday	515869	88955	39550	0.0767
Wednesday	590392	105035	46770	0.0792

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